4

Random matrices

We begin to study the non-asymptotic theory of random matrices, a study that will be continued in many further chapters. Section 4.1 is a quick reminder about singular values and matrix norms and their relationships. Section 4.2 introduces important geometric concepts – nets, covering and packing numbers, metric entropy, and discusses relations of these quantities with volume and coding. In Sections 4.4 and 4.6, we develop a basic *ε-net argument* and use it for random matrices. We first give a bound on the operator norm (Theorem 4.4.5) and then a stronger, two-sided bound on all singular values (Theorem 4.6.1) of random matrices. Three applications of random matrix theory are discussed in this chapter: a spectral clustering algorithm for recovering clusters, or communities, in complex networks (Section 4.5), covariance estimation (Section 4.7) and a spectral clustering algorithm for data presented as geometric point sets (Section 4.7.1).

4.1 Preliminaries on matrices

You should be familiar with the notion of singular value decomposition from a basic course in linear algebra; we recall it nevertheless. We will then introduce two matrix norms – operator and Frobenius, and discuss their relationships.

4.1.1 Singular value decomposition

The main object of our study will be an $m \times n$ matrix A with real entries. Recall that A can be represented using the *singular value decomposition* (SVD), which we can write as

$$A = \sum_{i=1}^{r} s_i u_i v_i^\mathsf{T}, \quad \text{where} \quad r = \operatorname{rank}(A). \tag{4.1}$$

Here the non-negative numbers $s_i = s_i(A)$ are called *singular values* of A, the vectors $u_i \in \mathbb{R}^m$ are called the *left singular vectors* of A, and the vectors $v_i \in \mathbb{R}^n$ are called the *right singular vectors* of A.

For convenience, we often extend the sequence of singular values by setting $s_i = 0$ for $r < i \le n$, and we arrange them in a non-increasing order:

$$s_1 \ge s_2 \ge \cdots \ge s_n \ge 0.$$

The left singular vectors u_i are the orthonormal eigenvectors of AA^T and the right singular vectors v_i are the orthonormal eigenvectors of $A^\mathsf{T}A$. The singular values s_i are the square roots of the eigenvalues λ_i of both AA^T and $A^\mathsf{T}A$:

$$s_i(A) = \sqrt{\lambda_i(AA^\mathsf{T})} = \sqrt{\lambda_i(A^\mathsf{T}A)}.$$

In particular, if A is a *symmetric* matrix, then the singular values of A are the absolute values of the eigenvalues λ_i of A:

$$s_i(A) = |\lambda_i(A)|,$$

and both left and right singular vectors of A are eigenvectors of A.

Courant-Fisher's *min-max theorem* offers the following variational characterization of eigenvalues $\lambda_i(A)$ of a symmetric matrix A, assuming they are arranged in a non-increasing order:

$$\lambda_i(A) = \max_{\dim E = i} \min_{x \in S(E)} \langle Ax, x \rangle, \tag{4.2}$$

Here the maximum is over all i-dimensional subspaces E of $\mathbb{R}^n$, and the minimum is over all unit vectors $x \in E$, and $S(E)$ denotes the unit Euclidean sphere in the subspace E. For the singular values, the min-max theorem immediately implies that

$$s_i(A) = \max_{\dim E = i} \min_{x \in S(E)} \|Ax\|_2.$$

Exercise 4.1.1. ☕ Suppose A is an invertible matrix with singular value decomposition

$$A = \sum_{i=1}^n s_i u_i v_i^\mathsf{T}.$$

Check that

$$A^{-1} = \sum_{i=1}^n \frac{1}{s_i} v_i u_i^\mathsf{T}.$$

4.1.2 Operator norm and the extreme singular values

The space of $m \times n$ matrices can be equipped with several classical norms. We mention two of them – operator and Frobenius norms – and emphasize their connection with the spectrum of A.

When we think of the space $\mathbb{R}^m$ along with the Euclidean norm $\|\cdot\|_2$ on it, we denote this Hilbert space ℓ_2^m. The matrix A acts as a linear operator from $\ell_2^n \to \ell_2^m$. Its *operator norm*, also called the *spectral norm*, is defined as

$$\|A\| := \|A : \ell_2^n \to \ell_2^m\| = \max_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Ax\|_2}{\|x\|_2} = \max_{x \in S^{n-1}} \|Ax\|_2.$$

Equivalently, the operator norm of A can be computed by maximizing the quadratic form $\langle Ax, y \rangle$ over all unit vectors x, y:

$$\|A\| = \max_{x \in S^{n-1},\, y \in S^{m-1}} \langle Ax, y \rangle.$$

In terms of spectrum, the operator norm of A equals the largest singular value of A:

$$s_1(A) = \|A\|.$$

(Check!)

The smallest singular value $s_n(A)$ also has a special meaning. By definition, it can only be non-zero for tall matrices where $m \geq n$. In this case, A has full rank n if and only if $s_n(A) > 0$. Moreover, $s_n(A)$ is a quantitative measure of *non-degeneracy* of A. Indeed, if A has full rank then

$$s_n(A) = \frac{1}{\|A^+\|}$$

where A^+ is the Moore-Penrose pseudoinverse of A. Its norm $\|A^+\|$ is the norm of the operator A^{-1} restricted to the image of A.

4.1.3 Frobenius norm

The *Frobenius norm*, also called *Hilbert-Schmidt* norm of a matrix A with entries A_{ij} is defined as

$$\|A\|_F = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} |A_{ij}|^2 \right)^{1/2}.$$

Thus the Frobenius norm is the Euclidean norm on the space of matrices $\mathbb{R}^{m\times n}$. In terms of singular values, the Frobenius norm can be computed as

$$\|A\|_F = \left(\sum_{i=1}^{r} s_i(A)^2 \right)^{1/2}.$$

The canonical inner product on $\mathbb{R}^{m\times n}$ can be represented in terms of matrices as

$$\langle A, B \rangle = \operatorname{tr}(A^{\mathsf{T}} B) = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} B_{ij}. \tag{4.3}$$

Obviously, the canonical inner product generates the canonical Euclidean norm, i.e.

$$\|A\|_F^2 = \langle A, A \rangle .$$

Let us now compare the operator and the Frobenius norm. If we look at the vector $s = (s_1, \ldots, s_r)$ of singular values of A, these norms become the ℓ_∞ and ℓ_2 norms, respectively:

$$\|A\| = \|s\|_\infty, \quad \|A\|_F = \|s\|_2.$$

Using the inequality $\|s\|_\infty \leq \|s\|_2 \leq \sqrt{r}\,\|s\|_\infty$ for $s \in \mathbb{R}^n$ (check it!) we obtain the best possible relation between the operator and Frobenius norms:

$$\|A\| \leq \|A\|_F \leq \sqrt{r}\,\|A\|. \tag{4.4}$$

Exercise 4.1.2. Prove the following bound on the singular values s_i of any matrix A:

$$s_i \le \frac{1}{\sqrt{i}} \|A\|_F.$$

4.1.4 Low-rank approximation

Suppose we want to approximate a given matrix A of rank r by a matrix A_k that has a given lower rank, say rank $k < r$. What is the best choice for A_k? In other words, what matrix A_k of rank k minimizes the distance to A? The distance can be measured by the operator norm or Frobenius norm.

In either case, *Eckart-Young-Mirsky's theorem* gives the answer to the low-rank approximation problem. It states that the minimizer A_k is obtained by truncating the singular value decomposition of A at the k-th term:

$$A_k = \sum_{i=1}^{k} s_i u_i v_i^\mathsf{T}.$$

In other words, the Eckart-Young-Mirsky theorem states that

$$\|A - A_k\| = \min_{\operatorname{rank}(A') \le k} \|A - A'\|.$$

A similar statement holds for the Frobenius norm (and, in fact, for any unitary-invariant norm). The matrix A_k is often called the *best rank k approximation* of A.

Exercise 4.1.3 (Best rank k approximation)**.** Let A_k be the best rank k approximation of a matrix A. Express $\|A - A_k\|^2$ and $\|A - A_k\|_F^2$ in terms of the singular values s_i of A.

4.1.5 Approximate isometries

The extreme singular values $s_1(A)$ and $s_n(A)$ have an important geometric meaning. They are respectively the smallest number M and the largest number m that make the following inequality true:

$$m\|x\|_2 \le \|Ax\|_2 \le M\|x\|_2 \quad \text{for all } x \in \mathbb{R}^n. \tag{4.5}$$

(Check!) Applying this inequality for $x - y$ instead of x and with the best bounds, we can rewrite it as

$$s_n(A)\|x - y\|_2 \le \|Ax - Ay\|_2 \le s_1(A)\|x - y\|_2 \quad \text{for all } x \in \mathbb{R}^n.$$

This means that the matrix A, acting as an operator from $\mathbb{R}^n$ to $\mathbb{R}^m$, can only change the distance between any points by a factor that lies between $s_n(A)$ and $s_1(A)$. Thus the extreme singular values control the *distortion* of the geometry of $\mathbb{R}^n$ under the action of A.

The best possible matrices in this sense, which preserve distances exactly, are called *isometries*. Let us recall their characterization, which can be proved using elementary linear algebra.

Exercise 4.1.4 (Isometries)**.** Let A be an $m \times n$ matrix with $m \geq n$. Prove that the following statements are equivalent.

(a) $A^\mathsf{T} A = I_n$.
(b) $P := AA^\mathsf{T}$ is an *orthogonal projection*[1] in $\mathbb{R}^m$ onto a subspace of dimension n.
(c) A is an *isometry*, or isometric embedding of $\mathbb{R}^n$ into $\mathbb{R}^m$, which means that

$$\|Ax\|_2 = \|x\|_2 \quad \text{for all } x \in \mathbb{R}^n.$$

(d) All singular values of A equal 1; equivalently

$$s_n(A) = s_1(A) = 1.$$

Quite often the conditions of Exercise 4.1.4 hold only approximately, in which case we regard A as an *approximate isometry*.

Lemma 4.1.5 (Approximate isometries)**.** *Let A be an $m \times n$ matrix and $\delta > 0$. Suppose that*

$$\|A^\mathsf{T} A - I_n\| \leq \max(\delta, \delta^2).$$

Then

$$(1-\delta)\|x\|_2 \leq \|Ax\|_2 \leq (1+\delta)\|x\|_2 \quad \textit{for all } x \in \mathbb{R}^n. \tag{4.6}$$

Consequently, all singular values of A are between $1-\delta$ and $1+\delta$:

$$1 - \delta \leq s_n(A) \leq s_1(A) \leq 1 + \delta. \tag{4.7}$$

Proof To prove (4.6), we may assume without loss of generality that $\|x\|_2 = 1$. (Why?) Then, using the assumption, we get

$$\max(\delta, \delta^2) \geq \left| \left\langle (A^\mathsf{T} A - I_n)x, x \right\rangle \right| = \left| \|Ax\|_2^2 - 1 \right|.$$

Applying the elementary inequality

$$\max(|z-1|, |z-1|^2) \leq |z^2 - 1|, \quad z \geq 0 \tag{4.8}$$

for $z = \|Ax\|_2$, we conclude that

$$\left| \|Ax\|_2 - 1 \right| \leq \delta.$$

This proves (4.6), which in turn implies (4.7) as we saw in the beginning of this section. □

Exercise 4.1.6 (Approximate isometries)**.** Prove the following converse to Lemma 4.1.5: if (4.7) holds, then

$$\|A^\mathsf{T} A - I_n\| \leq 3\max(\delta, \delta^2).$$

[1] Recall that P is a projection if $P^2 = P$, and P is called orthogonal if the image and kernel of P are orthogonal subspaces.

Remark 4.1.7 (Projections vs. isometries)**.** Consider an $n \times m$ matrix Q. Then

$$QQ^{\mathsf{T}} = I_n$$

if and only if

$$P := Q^{\mathsf{T}} Q$$

is an orthogonal projection in $\mathbb{R}^m$ onto a subspace of dimension n. (This can be checked directly or deduced from Exercise 4.1.4 by taking $A = Q^{\mathsf{T}}$.) In case this happens, the matrix Q itself is often called a *projection* from $\mathbb{R}^m$ onto $\mathbb{R}^n$.

Note that A is an isometric embedding of $\mathbb{R}^n$ into $\mathbb{R}^m$ if and only if A^{T} is a projection from $\mathbb{R}^m$ onto $\mathbb{R}^n$. These remarks can be also made for an approximate isometry A; the transpose A^{T} in this case is an *approximate projection*.

Exercise 4.1.8 (Isometries and projections from unitary matrices)**.** ☕ Canonical examples of isometries and projections can be constructed from a fixed unitary matrix U. Check that any sub-matrix of U obtained by selecting a subset of columns is an isometry, and any sub-matrix obtained by selecting a subset of rows is a projection.

4.2 Nets, covering numbers and packing numbers

We are going to develop a simple but powerful method – an ε-net argument – and illustrate its usefulness for the analysis of random matrices. In this section, we recall the concept of an *ε-net*, which you may have seen in a course in real analysis, and we relate it to some other basic notions – covering, packing, entropy, volume, and coding.

Definition 4.2.1 (ε-net)**.** Let (T, d) be a metric space. Consider a subset $K \subset T$ and let $\varepsilon > 0$. A subset $\mathcal{N} \subseteq K$ is called an *ε-net of K* if every point in K is within distance ε of some point of $\mathcal{N}$, i.e.

$$\forall x \in K \ \exists x_0 \in \mathcal{N} : \ d(x, x_0) \le \varepsilon.$$

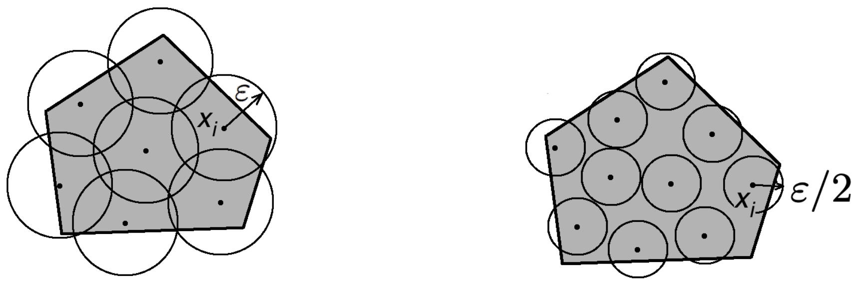
Equivalently, $\mathcal{N}$ is an ε-net of K if and only if K can be covered by balls with centers in $\mathcal{N}$ and radii ε, see Figure 4.1a.

If you ever feel confused by too much generality, it might be helpful to keep in mind an important example. Let $T = \mathbb{R}^n$ with d being the Euclidean distance, i.e.

$$d(x, y) = \|x - y\|_2, \quad x, y \in \mathbb{R}^n. \tag{4.9}$$

In this case, we cover a subset $K \subset \mathbb{R}^n$ by *round balls*, as shown in Figure 4.1a. We already saw an example of such covering in Corollary 0.0.4 where K was a polytope.

Definition 4.2.2 (Covering numbers)**.** The smallest possible cardinality of an ε-net of K is called the *covering number* of K and is denoted $\mathcal{N}(K, d, \varepsilon)$. Equivalently, $\mathcal{N}(K, d, \varepsilon)$ is the smallest number of closed balls with centers in K and radii ε whose union covers K.



(a) This covering of a pentagon K by seven ε-balls shows that $\mathcal{N}(K, \varepsilon) \le 7$.

(b) This packing of a pentagon K by ten $\varepsilon/2$-balls shows that $\mathcal{P}(K, \varepsilon) \ge 10$.

Figure 4.1 Packing and covering

Remark 4.2.3 (Compactness)**.** An important result in real analysis states that a subset K of a complete metric space (T, d) is *precompact* (i.e. the closure of K is compact) if and only if

$$\mathcal{N}(K, d, \varepsilon) < \infty \quad \text{for every } \varepsilon > 0.$$

Thus we can think about the magnitude $\mathcal{N}(K, d, \varepsilon)$ as a quantitative measure of compactness of K.

Closely related to covering is the notion of *packing*.

Definition 4.2.4 (Packing numbers)**.** A subset $\mathcal{N}$ of a metric space (T, d) is *ε-separated* if $d(x, y) > \varepsilon$ for all distinct points $x, y \in \mathcal{N}$. The largest possible cardinality of an ε-separated subset of a given set $K \subset T$ is called the *packing number* of K and is denoted $\mathcal{P}(K, d, \varepsilon)$.

Exercise 4.2.5 (Packing the balls into K)**.** ☕☕

(a) Suppose T is a normed space. Prove that $\mathcal{P}(K, d, \varepsilon)$ is the largest number of closed disjoint balls with centers in K and radii $\varepsilon/2$. See Figure 4.1b for an illustration.

(b) Show by example that the previous statement may be false for a general metric space T.

Lemma 4.2.6 (Nets from separated sets)**.** *Let $\mathcal{N}$ be a* maximal[2] *ε-separated subset of K. Then $\mathcal{N}$ is an ε-net of K.*

Proof Let $x \in K$; we want to show that there exists $x_0 \in \mathcal{N}$ such that $d(x, x_0) \le \varepsilon$. If $x \in \mathcal{N}$, the conclusion is trivial by choosing $x_0 = x$. Suppose now $x \notin \mathcal{N}$. The maximality assumption implies that $\mathcal{N} \cup \{x\}$ is not ε-separated. But this means precisely that $d(x, x_0) \le \varepsilon$ for some $x_0 \in \mathcal{N}$. □

Remark 4.2.7 (Constructing a net)**.** Lemma 4.2.6 leads to the following simple algorithm for constructing an ε-net of a given set K. Choose a point $x_1 \in K$

[2] Here by "maximal" we mean that adding any new point to $\mathcal{N}$ destroys the separation property.

arbitrarily, choose a point $x_2 \in K$ which is farther than ε from x_1, choose x_3 so that it is farther than ε from both x_1 and x_2, and so on. If K is compact, the algorithm terminates in finite time (why?) and gives an ε-net of K.

The covering and packing numbers are essentially equivalent:

Lemma 4.2.8 (Equivalence of covering and packing numbers)**.** *For any set $K \subset T$ and any $\varepsilon > 0$, we have*

$$\mathcal{P}(K, d, 2\varepsilon) \leq \mathcal{N}(K, d, \varepsilon) \leq \mathcal{P}(K, d, \varepsilon).$$

Proof The upper bound follows from Lemma 4.2.6. (How?)

To prove the lower bound, choose an 2ε-separated subset $\mathcal{P} = \{x_i\}$ in K and an ε-net $\mathcal{N} = \{y_j\}$ of K. By the definition of a net, each point x_i belongs a closed ε-ball centered at some point y_j. Moreover, since any closed ε-ball can not contain a pair of 2ε-separated points, each ε-ball centered at y_j may contain at most one point x_i. The pigeonhole principle then yields $|\mathcal{P}| \leq |\mathcal{N}|$. Since this happens for arbitrary packing $\mathcal{P}$ and covering $\mathcal{N}$, the lower bound in the lemma is proved. □

Exercise 4.2.9 (Allowing the centers to be outside K)**.** ☕☕☕ In our definition of the covering numbers of K, we required that the centers x_i of the balls $B(x_i, \varepsilon)$ that form a covering lie in K. Relaxing this condition, define the *exterior covering number* $\mathcal{N}^{\text{ext}}(K, d, \varepsilon)$ similarly but without requiring that $x_i \in K$. Prove that

$$\mathcal{N}^{\text{ext}}(K, d, \varepsilon) \leq \mathcal{N}(K, d, \varepsilon) \leq \mathcal{N}^{\text{ext}}(K, d, \varepsilon/2).$$

Exercise 4.2.10 (Monotonicity)**.** ☕☕☕ Give a counterexample to the following monotonicity property:

$$L \subset K \quad \text{implies} \quad \mathcal{N}(L, d, \varepsilon) \leq \mathcal{N}(K, d, \varepsilon).$$

Prove an approximate version of monotonicity:

$$L \subset K \quad \text{implies} \quad \mathcal{N}(L, d, \varepsilon) \leq \mathcal{N}(K, d, \varepsilon/2).$$

4.2.1 Covering numbers and volume

Let us now specialize our study of covering numbers to the most important example where $T = \mathbb{R}^n$ with the Euclidean metric

$$d(x, y) = \|x - y\|_2$$

as in (4.9). To ease the notation, we often omit the metric when it is understood, thus writing

$$\mathcal{N}(K, \varepsilon) = \mathcal{N}(K, d, \varepsilon).$$

If the covering numbers measure the size of K, how are they related to the most classical measure of size, the volume of K in $\mathbb{R}^n$? There could not be a full equivalence between these two quantities, since "flat" sets have zero volume but non-zero covering numbers.

Still, there is a useful partial equivalence, which is often quite sharp. It is based on the notion of *Minkowski sum* of sets in $\mathbb{R}^n$.

Definition 4.2.11 (Minkowski sum)**.** Let A and B be subsets of $\mathbb{R}^n$. The *Minkowski sum* $A + B$ is defined as

$$A + B := \{a + b : \; a \in A, \; b \in B\}.$$

Figure 4.2 shows an example of Minkowski sum of two sets on the plane.

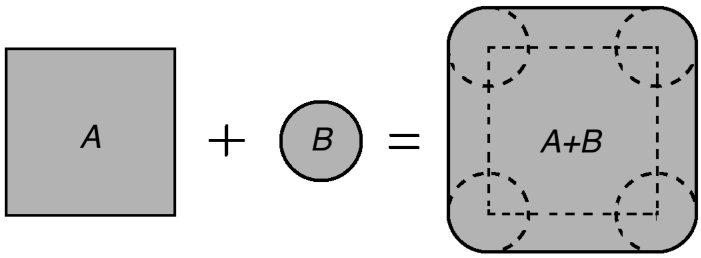


Figure 4.2 Minkowski sum of a square and a circle is a square with rounded corners.

Proposition 4.2.12 (Covering numbers and volume)**.** *Let K be a subset of $\mathbb{R}^n$ and $\varepsilon > 0$. Then*

$$\frac{|K|}{|\varepsilon B_2^n|} \leq \mathcal{N}(K, \varepsilon) \leq \mathcal{P}(K, \varepsilon) \leq \frac{|(K + (\varepsilon/2)B_2^n)|}{|(\varepsilon/2)B_2^n|}.$$

Here $|\cdot|$ denotes the volume in $\mathbb{R}^n$, B_2^n denotes the unit Euclidean ball[3] *in $\mathbb{R}^n$, so εB_2^n is a Euclidean ball with radius ε.*

Proof The middle inequality follows from Lemma 4.2.8, so all we need to prove is the left and right bounds.

(Lower bound) Let $N := \mathcal{N}(K, \varepsilon)$. Then K can be covered by N balls with radii ε. Comparing the volumes, we obtain

$$|K| \leq N \cdot |\varepsilon B_2^n|,$$

Dividing both sides by $|\varepsilon B_2^n|$ yields the lower bound.

(Upper bound) Let $N := \mathcal{P}(K, \varepsilon)$. Then one can construct N closed disjoint balls $B(x_i, \varepsilon/2)$ with centers $x_i \in K$ and radii $\varepsilon/2$ (see Exercise 4.2.5). While these balls may not need to fit entirely in K (see Figure 4.1b), they do fit in a slightly inflated set, namely $K + (\varepsilon/2)B_2^n$. (Why?) Comparing the volumes, we obtain

$$N \cdot |(\varepsilon/2)B_2^n| \leq |K + (\varepsilon/2)B_2^n|.$$

which leads to the upper bound in the proposition. □

[3] Thus $B_2^n = \{x \in \mathbb{R}^n : \; \|x\|_2 \leq 1\}$.

An important consequence of the volumetric bound (4.10) is that the covering (and thus packing) numbers of the Euclidean ball, as well as many other sets, are *exponential* in the dimension n. Let us check this.

Corollary 4.2.13 (Covering numbers of the Euclidean ball)**.** *The covering numbers of the unit Euclidean ball B_2^n satisfy the following for any $\varepsilon > 0$:*

$$\left(\frac{1}{\varepsilon}\right)^n \le \mathcal{N}(B_2^n, \varepsilon) \le \left(\frac{2}{\varepsilon} + 1\right)^n.$$

The same upper bound is true for the unit Euclidean sphere S^{n-1}.

Proof The lower bound follows immediately from Proposition 4.2.12, since the volume in $\mathbb{R}^n$ scales as

$$|\varepsilon B_2^n| = \varepsilon^n \, |B_2^n|.$$

The upper bound follows from Proposition 4.2.12, too:

$$\mathcal{N}(B_2^n, \varepsilon) \le \frac{|(1+\varepsilon/2)B_2^n|}{|(\varepsilon/2)B_2^n|} = \frac{(1+\varepsilon/2)^n}{(\varepsilon/2)^n} = \left(\frac{2}{\varepsilon} + 1\right)^n.$$

The upper bound for the sphere can be proved in the same way. □

To simplify the bound a bit, note that in the non-trivial range $\varepsilon \in (0, 1]$ we have

$$\left(\frac{1}{\varepsilon}\right)^n \le \mathcal{N}(B_2^n, \varepsilon) \le \left(\frac{3}{\varepsilon}\right)^n. \tag{4.10}$$

In the trivial range where $\varepsilon > 1$, the unit ball can be covered by just one ε-ball, so $N(B_2^n, \varepsilon) = 1$.

The volumetric argument we just gave works well in many other situations. Let us give an important example.

Definition 4.2.14 (Hamming cube)**.** The Hamming cube $\{0,1\}^n$ consists of all binary strings of length n. The *Hamming distance* $d_H(x, y)$ between two binary strings is defined as the number of bits where x and y disagree, i.e.

$$d_H(x, y) := \#\Big\{ i : \; x(i) \ne y(i) \Big\}, \quad x, y \in \{0,1\}^n.$$

Endowed with this metric, the Hamming cube is a metric space $(\{0,1\}^n, d_H)$, which is sometimes called the *Hamming space*.

Exercise 4.2.15. ☕ Check that d_H is indeed a metric.

Exercise 4.2.16 (Covering and packing numbers of the Hamming cube)**.** ☕☕☕ Let $K = \{0,1\}^n$. Prove that for every integer $m \in [0, n]$, we have

$$\frac{2^n}{\sum_{k=0}^{m} \binom{n}{k}} \le \mathcal{N}(K, d_H, m) \le \mathcal{P}(K, d_H, m) \le \frac{2^n}{\sum_{k=0}^{\lfloor m/2 \rfloor} \binom{n}{k}}$$

Hint: Adapt the volumetric argument by replacing volume by cardinality.

To make these bounds easier to compute, one can use the bounds for binomial sums from Exercise 0.0.5.

4.3 Application: error correcting codes

Covering and packing arguments frequently appear in applications to *coding theory*. Here we give two examples that relate covering and packing numbers to complexity and error correction.

4.3.1 Metric entropy and complexity

Intuitively, the covering and packing numbers measure the *complexity* of a set K. The logarithm of the covering numbers $\log_2 \mathcal{N}(K, \varepsilon)$ is often called the *metric entropy* of K. As we will see now, the metric entropy is equivalent to the number of bits needed to encode points in K.

Proposition 4.3.1 (Metric entropy and coding). *Let (T, d) be a metric space, and consider a subset $K \subset T$. Let $\mathcal{C}(K, d, \varepsilon)$ denote the smallest number of bits sufficient to specify every point $x \in K$ with accuracy ε in the metric d. Then*

$$\log_2 \mathcal{N}(K, d, \varepsilon) \le \mathcal{C}(K, d, \varepsilon) \le \left\lceil \log_2 \mathcal{N}(K, d, \varepsilon/2) \right\rceil .$$

Proof **(Lower bound)** Assume $\mathcal{C}(K, d, \varepsilon) \le N$. This means that there exists a transformation ("encoding") of points $x \in K$ into bit strings of length N, which specifies every point with accuracy ε. Such a transformation induces a partition of K into at most 2^N subsets, which are obtained by grouping the points represented by the same bit string; see Figure 4.3 for an illustration. Each subset must have diameter[4] at most ε, and thus it can be covered by a ball centered in K and with radius ε. (Why?) Thus K can be covered by at most 2^N balls with radii ε. This implies that $\mathcal{N}(K, d, \varepsilon) \le 2^N$. Taking logarithms on both sides, we obtain the lower bound in the proposition.

(Upper bound) Assume $\log_2 \mathcal{N}(K, d, \varepsilon/2) \le N$ for some integer N. This means that there exists an $(\varepsilon/2)$-net $\mathcal{N}$ of K with cardinality $|\mathcal{N}| \le 2^N$. To every point $x \in K$, let us assign a point $x_0 \in \mathcal{N}$ that is closest to x. Since there are at most 2^N such points, N bits are sufficient to specify the point x_0. It remains to note that the encoding $x \mapsto x_0$ represents points in K with accuracy ε. Indeed, if both x and y are encoded by the same x_0 then, by triangle inequality,

$$d(x, y) \le d(x, x_0) + d(y, x_0) \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

This shows that $\mathcal{C}(K, d, \varepsilon) \le N$. This completes the proof. □

4.3.2 Error correcting codes

Suppose Alice wants to send Bob a message that consists of k letters, such as

$$x := \text{“}\textit{fill the glass}\text{”}.$$

[4] If (T, d) is a metric space and $K \subset T$, the diameter of the set K is defined as $\operatorname{diam}(K) := \sup\{d(x, y) : x, y \in K\}$.

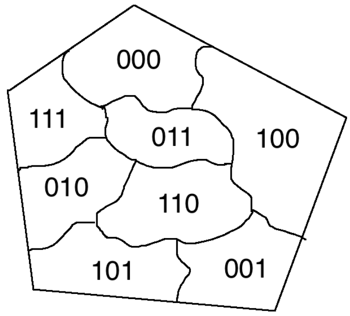


Figure 4.3 Encoding points in K as N-bit strings induces a partition of K into at most 2^N subsets.

Suppose further that an adversary may corrupt Alice's message by changing at most r letters in it. For example, Bob may receive

$$y := \text{"\textit{bill the class}"}$$

if $r = 2$. Is there a way to protect the communication channel between Alice and Bob, a method that can correct adversarial errors?

A common approach relies on using *redundancy.* Alice would encode her k-letter message into a longer, n-letter, message for some $n > k$, hoping that the extra information would help Bob get her message right despite any r errors.

Example 4.3.2 (Repetition code)**.** Alice may just repeat her message several times, thus sending to Bob

$$E(x) := \text{"\textit{fill the glass fill the glass fill the glass fill the glass fill the glass}"}.$$

Bob could then use the *majority decoding*: to determine the value of any particular letter, he would look at the received copies of it in $E(x)$ and choose the value that occurs most frequently. If the original message x is repeated $2r + 1$ times, then the majority decoding recovers x exactly even when r letters of $E(x)$ are corrupted. (Why?)

The problem with majority decoding is that it is very inefficient: it uses

$$n = (2r + 1)k \tag{4.11}$$

letters to encode a k-letter message. As we will see shortly, there exist error correcting codes with much smaller n.

But first let us formalize the notion of an error correcting code – an encoding of k-letter strings into n-letter strings that can correct r errors. For convenience, instead of using the English alphabet we shall work with the binary alphabet consisting of two letters 0 and 1.

Definition 4.3.3 (Error correcting code)**.** Fix integers k, n and r. Two maps

$$E : \{0,1\}^k \to \{0,1\}^n \quad \text{and} \quad D : \{0,1\}^n \to \{0,1\}^k$$

are called *encoding* and *decoding* maps that can correct r errors if we have

$$D(y) = x$$

for every word $x \in \{0,1\}^k$ and every string $y \in \{0,1\}^n$ that differs from $E(x)$ in at most r bits. The encoding map E is called an *error correcting code*; its image $E(\{0,1\}^k)$ is called a *codebook* (and very often the image itself is called the *error correcting code*); the elements $E(x)$ of the image are called *codewords*.

We now relate error correction to packing numbers of the Hamming cube $(\{0,1\}^n, d_H)$ where d_H is the Hamming metric we introduced in Definition 4.2.14.

Lemma 4.3.4 (Error correction and packing)**.** *Assume that positive integers k, n and r are such that*

$$\log_2 \mathcal{P}(\{0,1\}^n, d_H, 2r) \geq k.$$

Then there exists an error correcting code that encodes k-bit strings into n-bit strings and can correct r errors.

Proof By assumption, there exists a subset $\mathcal{N} \subset \{0,1\}^n$ with cardinality $|\mathcal{N}| = 2^k$ and such that the closed balls centered at the points in $\mathcal{N}$ and with radii r are disjoint. (Why?) We then define the encoding and decoding maps as follows: choose $E : \{0,1\}^k \to \mathcal{N}$ to be an arbitrary one-to-one map and $D : \{0,1\}^n \to \{0,1\}^k$ to be a nearest neighbor decoder.[5]

Now, if $y \in \{0,1\}^n$ differs from $E(x)$ in at most r bits, y lies in the closed ball centered at $E(x)$ and with radius r. Since such balls are disjoint by construction, y must be strictly closer to $E(x)$ than to any other codeword $E(x')$ in $\mathcal{N}$. Thus the nearest-neighbor decoding decodes y correctly, i.e. $D(y) = x$. This completes the proof. □

Let us substitute into Lemma 4.3.4 the bounds on the packing numbers of the Hamming cube from Exercise 4.2.16.

Theorem 4.3.5 (Guarantees for an error correcting code)**.** *Assume that positive integers k, n and r are such that*

$$n \geq k + 2r \log_2 \left(\frac{en}{2r} \right).$$

Then there exists an error correcting code that encodes k-bit strings into n-bit strings and can correct r errors.

Proof Passing from packing to covering numbers using Lemma 4.2.8 and then using the bounds on the covering numbers from Exercises 4.2.16 (and simplifying using Exercise 0.0.5), we get

$$\mathcal{P}(\{0,1\}^n, d_H, 2r) \geq \mathcal{N}(\{0,1\}^n, d_H, 2r) \geq 2^n \left(\frac{2r}{en} \right)^{2r}.$$

[5] Formally, we set $D(y) = x_0$ where $E(x_0)$ is the closest codeword in $\mathcal{N}$ to y; break ties arbitrarily.

By assumption, this quantity is further bounded below by 2^k. An application of Lemma 4.3.4 completes the proof. $\square$

Informally, Theorem 4.3.5 shows that we can correct r errors if we make the information overhead $n-k$ almost linear in r:

$$n - k \asymp r \log\left(\frac{n}{r}\right).$$

This overhead is much smaller than for the repetition code (4.11). For example, to correct two errors in Alice's twelve-letter message *"fill the glass"*, encoding it into a 30-letter codeword would suffice.

Remark 4.3.6 (Rate)**.** The guarantees of a given error correcting code are traditionally expressed in terms of the tradeoff between the *rate* and *fraction of errors*, defined as

$$R := \frac{k}{n} \quad \text{and} \quad \delta := \frac{r}{n}.$$

Theorem 4.3.5 states that there exist error correcting codes with rate as high as

$$R \geq 1 - f(2\delta)$$

where $f(t) = t \log_2(e/t)$.

Exercise 4.3.7 (Optimality)**.** ☕☕☕

(a) Prove the converse to the statement of Lemma 4.3.4.
(b) Deduce a converse to Theorem 4.3.5. Conclude that for any error correcting code that encodes k-bit strings into n-bit strings and can correct r errors, the rate must be

$$R \leq 1 - f(\delta)$$

where $f(t) = t \log_2(1/t)$ as before.

4.4 Upper bounds on random sub-gaussian matrices

We are now ready to begin to study the non-asymptotic theory of random matrices. Random matrix theory is concerned with $m \times n$ matrices A with random entries. The central questions of this theory are about the distributions of singular values, eigenvalues (if A is symmetric) and eigenvectors of A.

Theorem 4.4.5 will give a first bound on the operator norm (equivalently, on the largest singular value) of a random matrix with independent sub-gaussian entries. It is neither the sharpest nor the most general result; it will be sharpened and extended in Sections 4.6 and 6.5.

But before we do this, let us pause to learn how ε-nets can help us compute the operator norm of a matrix.

4.4.1 Computing the norm on a net

The notion of ε-nets can help us to simplify various problems involving high-dimensional sets. One such problem is the computation of the operator norm of an $m \times n$ matrix A. The operator norm was defined in Section 4.1.2 as

$$\|A\| = \max_{x \in S^{n-1}} \|Ax\|_2.$$

Thus, to evaluate $\|A\|$ one needs to control $\|Ax\|$ uniformly over the sphere S^{n-1}. We will show that instead of the entire sphere, it is enough to gain control just over an ε-net of the sphere (in the Euclidean metric).

Lemma 4.4.1 (Computing the operator norm on a net)**.** *Let A be an $m \times n$ matrix and $\varepsilon \in [0, 1)$. Then, for any ε-net $\mathcal{N}$ of the sphere S^{n-1}, we have*

$$\sup_{x \in \mathcal{N}} \|Ax\|_2 \le \|A\| \le \frac{1}{1-\varepsilon} \cdot \sup_{x \in \mathcal{N}} \|Ax\|_2$$

Proof The lower bound in the conclusion is trivial since $\mathcal{N} \subset S^{n-1}$. To prove the upper bound, fix a vector $x \in S^{n-1}$ for which

$$\|A\| = \|Ax\|_2$$

and choose $x_0 \in \mathcal{N}$ that approximates x so that

$$\|x - x_0\|_2 \le \varepsilon.$$

By the definition of the operator norm, this implies

$$\|Ax - Ax_0\|_2 = \|A(x - x_0)\|_2 \le \|A\| \|x - x_0\|_2 \le \varepsilon \|A\|.$$

Using triangle inequality, we find that

$$\|Ax_0\|_2 \ge \|Ax\|_2 - \|Ax - Ax_0\|_2 \ge \|A\| - \varepsilon\|A\| = (1-\varepsilon)\|A\|.$$

Dividing both sides of this inequality by $1 - \varepsilon$, we complete the proof. □

Exercise 4.4.2. ☕ Let $x \in \mathbb{R}^n$ and $\mathcal{N}$ be an ε-net of the sphere S^{n-1}. Show that

$$\sup_{y \in \mathcal{N}} \langle x, y \rangle \le \|x\|_2 \le \frac{1}{1-\varepsilon} \sup_{y \in \mathcal{N}} \langle x, y \rangle .$$

Recall from Section 4.1.2 that the operator norm of A can be computed by maximizing a quadratic form:

$$\|A\| = \max_{x \in S^{n-1},\, y \in S^{m-1}} \langle Ax, y \rangle .$$

Moreover, for symmetric matrices one can take $x = y$ in this formula. The following exercise shows that instead of controlling the quadratic form on the spheres, it suffices to have control just over the ε-nets.

Exercise 4.4.3 (Quadratic form on a net)**.** ☕☕ Let A be an $m \times n$ matrix and $\varepsilon \in [0, 1/2)$.

(a) Show that for any ε-net $\mathcal{N}$ of the sphere S^{n-1} and any ε-net $\mathcal{M}$ of the sphere S^{m-1}, we have

$$\sup_{x\in\mathcal{N},\, y\in\mathcal{M}} \langle Ax, y\rangle \le \|A\| \le \frac{1}{1-2\varepsilon}\cdot \sup_{x\in\mathcal{N},\, y\in\mathcal{M}} \langle Ax, y\rangle .$$

(b) Moreover, if $m = n$ and A is symmetric, show that

$$\sup_{x\in\mathcal{N}} |\langle Ax, x\rangle| \le \|A\| \le \frac{1}{1-2\varepsilon}\cdot \sup_{x\in\mathcal{N}} |\langle Ax, x\rangle|.$$

Hint: Proceed similarly to the proof of Lemma 4.4.1 and use the identity $\langle Ax, y\rangle - \langle Ax_0, y_0\rangle = \langle Ax, y - y_0\rangle + \langle A(x - x_0), y_0\rangle$.

Exercise 4.4.4 (Deviation of the norm on a net)**.** ☕☕☕ Let A be an $m \times n$ matrix, $\mu \in \mathbb{R}$ and $\varepsilon \in [0, 1/2)$. Show that for any ε-net $\mathcal{N}$ of the sphere S^{n-1}, we have

$$\sup_{x\in S^{n-1}} \big|\|Ax\|_2 - \mu\big| \le \frac{C}{1-2\varepsilon}\cdot \sup_{x\in\mathcal{N}} \big|\|Ax\|_2 - \mu\big| .$$

Hint: Assume without loss of generality that $\mu = 1$. Represent $\|Ax\|_2^2 - 1$ as a quadratic form $\langle Rx, x\rangle$ where $R = A^{\mathsf{T}}A - I_n$. Use Exercise 4.4.3 to compute the maximum of this quadratic form on a net.

4.4.2 The norms of sub-gaussian random matrices

We are ready for the first result on random matrices. The following theorem states that the norm of an $m\times n$ random matrix A with independent sub-gaussian entries satisfies

$$\|A\| \lesssim \sqrt{m} + \sqrt{n}$$

with high probability.

Theorem 4.4.5 (Norm of matrices with sub-gaussian entries)**.** *Let A be an $m \times n$ random matrix whose entries A_{ij} are independent, mean zero, sub-gaussian random variables. Then, for any $t > 0$ we have*[6]

$$\|A\| \le CK\left(\sqrt{m} + \sqrt{n} + t\right)$$

with probability at least $1 - 2\exp(-t^2)$. Here $K = \max_{i,j} \|A_{ij}\|_{\psi_2}$.

Proof This proof is an example of an *ε-net argument*. We need to control $\langle Ax, y\rangle$ for all vectors x and y on the unit sphere. To this end, we will discretize the sphere using a net (approximation step), establish a tight control of $\langle Ax, y\rangle$ for fixed vectors x and y from the net (concentration step), and finish by taking a union bound over all x and y in the net.

Step 1: Approximation. Choose $\varepsilon = 1/4$. Using Corollary 4.2.13, we can find an ε-net $\mathcal{N}$ of the sphere S^{n-1} and ε-net $\mathcal{M}$ of the sphere S^{m-1} with cardinalities

$$|\mathcal{N}| \le 9^n \quad\text{and}\quad |\mathcal{M}| \le 9^m. \tag{4.12}$$

[6] In results like this, C and c will always denote some positive absolute constants.

By Exercise 4.4.3, the operator norm of A can be bounded using these nets as follows:

$$\|A\| \le 2 \max_{x\in\mathcal{N},\, y\in\mathcal{M}} \langle Ax, y\rangle . \tag{4.13}$$

Step 2: Concentration. Fix $x \in \mathcal{N}$ and $y \in \mathcal{M}$. Then the quadratic form

$$\langle Ax, y\rangle = \sum_{i=1}^{n}\sum_{j=1}^{m} A_{ij} x_i y_j$$

is a sum of independent, sub-gaussian random variables. Proposition 2.6.1 states that the sum is sub-gaussian, and

$$\begin{aligned}\big\|\langle Ax, y\rangle\big\|_{\psi_2}^2 &\le C\sum_{i=1}^{n}\sum_{j=1}^{m}\|A_{ij}x_iy_j\|_{\psi_2}^2 \le CK^2\sum_{i=1}^{n}\sum_{j=1}^{m} x_i^2 y_j^2\\ &= CK^2\Big(\sum_{i=1}^{n} x_i^2\Big)\Big(\sum_{j=1}^{m} y_i^2\Big) = CK^2.\end{aligned}$$

Recalling (2.14), we can restate this as the tail bound

$$\mathbb{P}\left\{\langle Ax, y\rangle \ge u\right\} \le 2\exp(-cu^2/K^2), \quad u \ge 0. \tag{4.14}$$

Step 3: Union bound. Next, we unfix x and y using a union bound. Suppose the event $\max_{x\in\mathcal{N},\, y\in\mathcal{M}} \langle Ax, y\rangle \ge u$ occurs. Then there exist $x \in \mathcal{N}$ and $y \in \mathcal{M}$ such that $\langle Ax, y\rangle \ge u$. Thus the union bound yields

$$\mathbb{P}\left\{\max_{x\in\mathcal{N},\, y\in\mathcal{M}} \langle Ax, y\rangle \ge u\right\} \le \sum_{x\in\mathcal{N},\, y\in\mathcal{M}} \mathbb{P}\left\{\langle Ax, y\rangle \ge u\right\}.$$

Using the tail bound (4.14) and the estimate (4.12) on the sizes of $\mathcal{N}$ and $\mathcal{M}$, we bound the probability above by

$$9^{n+m}\cdot 2\exp(-cu^2/K^2). \tag{4.15}$$

Choose

$$u = CK(\sqrt{n} + \sqrt{m} + t). \tag{4.16}$$

Then $u^2 \ge C^2K^2(n+m+t^2)$, and if the constant C is chosen sufficiently large, the exponent in (4.15) is large enough, say $cu^2/K^2 \ge 3(n+m)+t^2$. Thus

$$\mathbb{P}\left\{\max_{x\in\mathcal{N},\, y\in\mathcal{M}} \langle Ax, y\rangle \ge u\right\} \le 9^{n+m}\cdot 2\exp\left(-3(n+m)-t^2\right) \le 2\exp(-t^2).$$

Finally, combining this with (4.13), we conclude that

$$\mathbb{P}\left\{\|A\| \ge 2u\right\} \le 2\exp(-t^2).$$

Recalling our choice of u in (4.16), we complete the proof. □

Exercise 4.4.6 (Expected norm). ☕ Deduce from Theorem 4.4.5 that

$$\mathbb{E}\,\|A\| \le CK\left(\sqrt{m}+\sqrt{n}\right).$$

Exercise 4.4.7 (Optimality)**.** ☕☕ Suppose that in Theorem 4.4.5 the entries A_{ij} have unit variances. Prove that for sufficiently large n and m one has

$$\mathbb{E}\,\|A\| \geq \frac{1}{4}\left(\sqrt{m}+\sqrt{n}\right).$$

Hint: Bound the operator norm of A below by the Euclidean norm of the first column and first row; use the concentration of the norm (Exercise 3.1.4) to complete the proof.

Theorem 4.4.5 can be easily extended for symmetric matrices, and the bound for them is

$$\|A\| \lesssim \sqrt{n}$$

with high probability.

Corollary 4.4.8 (Norm of symmetric matrices with sub-gaussian entries)**.** *Let A be an $n \times n$ symmetric random matrix whose entries A_{ij} on and above the diagonal are independent, mean zero, sub-gaussian random variables. Then, for any $t > 0$ we have*

$$\|A\| \leq CK\left(\sqrt{n}+t\right)$$

with probability at least $1-4\exp(-t^2)$. Here $K = \max_{i,j} \|A_{ij}\|_{\psi_2}$.

Proof Decompose A into the upper-triangular part A^+ and lower-triangular part A^-. It does not matter where the diagonal goes; let us include it into A^+ to be specific. Then

$$A = A^+ + A^-.$$

Theorem 4.4.5 applies for each part A^+ and A^- separately. By a union bound, we have simultaneously

$$\|A^+\| \leq CK\left(\sqrt{n}+t\right) \quad \text{and} \quad \|A^-\| \leq CK\left(\sqrt{n}+t\right)$$

with probability at least $1-4\exp(-t^2)$. Since by the triangle inequality $\|A\| \leq \|A^+\| + \|A^-\|$, the proof is complete. □

4.5 Application: community detection in networks

Results of random matrix theory are useful in many applications. Here we give an illustration in the analysis of networks.

Real-world networks tend to have *communities* – clusters of tightly connected vertices. Finding the communities accurately and efficiently is one of the main problems in network analysis, known as the *community detection problem.*

4.5.1 Stochastic Block Model

We will try to solve the community detection problem for a basic probabilistic model of a network with two communities. It is a simple extension of the Erdös-Rényi model of random graphs, which we described in Section 2.4.

Definition 4.5.1 (Stochastic block model)**.** Divide n vertices into two sets ("communities") of sizes $n/2$ each. Construct a random graph G by connecting every pair of vertices independently with probability p if they belong to the same community and q if they belong to different communities. This distribution on graphs is called the *stochastic block model*[7] and is denoted $G(n, p, q)$.

In the partial case where $p = q$ we obtain the Erdös-Rényi model $G(n, p)$. But we assume that $p > q$ here. In this case, edges are more likely to occur within than across communities. This gives the network a community structure; see Figure 4.4.

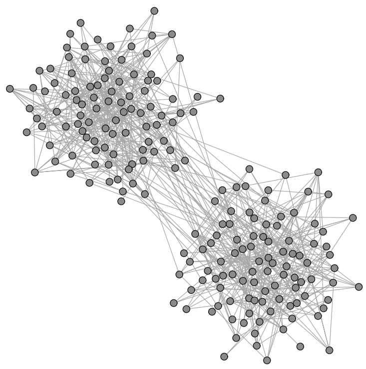


Figure 4.4 A random graph generated according to the stochastic block model $G(n, p, q)$ with $n = 200$, $p = 1/20$ and $q = 1/200$.

4.5.2 Expected adjacency matrix

It is convenient to identify a graph G with its adjacency matrix A which we introduced in Definition 3.6.2. For a random graph $G \sim G(n, p, q)$, the adjacency matrix A is a *random matrix*, and we will examine A using the tools we developed earlier in this chapter.

It is enlightening to split A into deterministic and random parts,

$$A = D + R,$$

where D is the expectation of A. We may think about D as an informative part (the "signal") and R as "noise".

To see why D is informative, let us compute its eigenstructure. The entries A_{ij} have a Bernoulli distribution; they are either $\mathrm{Ber}(p)$ or $\mathrm{Ber}(q)$ depending on the community membership of vertices i and j. Thus the entries of D are either p or q, depending on the membership. For illustration, if we group the vertices that belong to the same community together, then for $n = 4$ the matrix D will look

[7] The term *stochastic block model* can also refer to a more general model of random graphs with multiple communities of variable sizes.

like this:

$$D = \mathbb{E}\, A = \left[\begin{array}{cc|cc} p & p & q & q \\ p & p & q & q \\ \hline q & q & p & p \\ q & q & p & p \end{array}\right].$$

Exercise 4.5.2. ☕☕ Check that the matrix D has rank 2, and the non-zero eigenvalues λ_i and the corresponding eigenvectors u_i are

$$\lambda_1 = \left(\frac{p+q}{2}\right)n, \quad u_1 = \left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ \hline 1 \\ \vdots \\ 1 \end{array}\right]; \quad \lambda_2 = \left(\frac{p-q}{2}\right)n, \quad u_2 = \left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ \hline -1 \\ \vdots \\ -1 \end{array}\right]. \tag{4.17}$$

The important object here is the second eigenvector u_2. It contains all information about the community structure. If we knew u_2, we would identify the communities precisely based on the sizes of coefficients of u_2.

But we do not know $D = \mathbb{E}\, A$ and so we do not have access to u_2. Instead, we know $A = D + R$, a noisy version of D. The level of the signal D is

$$\|D\| = \lambda_1 \asymp n$$

while the level of the noise R can be estimated using Corollary 4.4.8:

$$\|R\| \le C\sqrt{n} \quad \text{with probability at least } 1 - 4e^{-n}. \tag{4.18}$$

Thus, for large n, the noise R is much smaller than the signal D. In other words, A is close to D, and thus we should be able to use A instead of D to extract the community information. This can be justified using the classical perturbation theory for matrices.

4.5.3 Perturbation theory

Perturbation theory describes how the eigenvalues and eigenvectors change under matrix perturbations. For the eigenvalues, we have

Theorem 4.5.3 (Weyl's inequality)**.** *For any symmetric matrices S and T with the same dimensions, we have*

$$\max_i |\lambda_i(S) - \lambda_i(T)| \le \|S - T\|.$$

Thus, the operator norm controls the stability of the spectrum.

Exercise 4.5.4. ☕☕ Deduce Weyl's inequality from the Courant-Fisher's min-max characterization of eigenvalues (4.2).

A similar result holds for eigenvectors, but we need to be careful to track the same eigenvector before and after perturbation. If the eigenvalues $\lambda_i(S)$ and $\lambda_{i+1}(S)$ are too close to each other, the perturbation can swap their order and force us to compare the wrong eigenvectors. To prevent this from happening, we can assume that the eigenvalues of S are well separated.

Theorem 4.5.5 (Davis-Kahan)**.** *Let S and T be symmetric matrices with the same dimensions. Fix i and assume that the i-th largest eigenvalue of S is well separated from the rest of the spectrum:*

$$\min_{j:j\neq i} |\lambda_i(S) - \lambda_j(S)| = \delta > 0.$$

Then the angle between the eigenvectors of S and T corresponding to the i-th largest eigenvalues (as a number between 0 and $\pi/2$) satisfies

$$\sin\angle\left(v_i(S),\, v_i(T)\right) \leq \frac{2\|S-T\|}{\delta}.$$

We do not prove the Davis-Kahan theorem here.

The conclusion of the Davis-Kahan theorem implies that the *unit* eigenvectors $v_i(S)$ and $v_i(T)$ are close to each other up to a sign, namely

$$\exists \theta \in \{-1,1\}: \quad \|v_i(S) - \theta v_i(T)\|_2 \leq \frac{2^{3/2}\|S-T\|}{\delta}. \tag{4.19}$$

(Check!)

4.5.4 Spectral Clustering

Returning to the community detection problem, let us apply the Davis-Kahan Theorem for $S = D$ and $T = A = D + R$, and for the second largest eigenvalue. We need to check that λ_2 is well separated from the rest of the spectrum of D, that is from 0 and λ_1. The distance is

$$\delta = \min(\lambda_2,\, \lambda_1 - \lambda_2) = \min\left(\frac{p-q}{2},\, q\right) n =: \mu n.$$

Recalling the bound (4.18) on $R = T - S$ and applying (4.19), we can bound the distance between the unit eigenvectors of D and A. It follows that there exists a sign $\theta \in \{-1,1\}$ such that

$$\|v_2(D) - \theta v_2(A)\|_2 \leq \frac{C\sqrt{n}}{\mu n} = \frac{C}{\mu\sqrt{n}}$$

with probability at least $1 - 4e^{-n}$. We already computed the eigenvectors $u_i(D)$ of D in (4.17), but there they had norm $\sqrt{n}$. So, multiplying both sides by $\sqrt{n}$, we obtain in this normalization that

$$\|u_2(D) - \theta u_2(A)\|_2 \leq \frac{C}{\mu}.$$

It follows that that the *signs* of most coefficients of $\theta v_2(A)$ and $v_2(D)$ must agree. Indeed, we know that

$$\sum_{j=1}^{n} |u_2(D)_j - \theta u_2(A)_j|^2 \le \frac{C}{\mu^2}. \tag{4.20}$$

and we also know from (4.17) that the coefficients $u_2(D)_j$ are all ± 1. So, every coefficient j on which the signs of $\theta v_2(A)_j$ and $v_2(D)_j$ disagree contributes at least 1 to the sum in (4.20). Thus the number of disagreeing signs must be bounded by

$$\frac{C}{\mu^2}.$$

Summarizing, we can use the vector $v_2(A)$ to accurately estimate the vector $v_2 = v_2(D)$ in (4.17), whose signs identify the two communities. This method for community detection is usually called *spectral clustering.* Let us explicitly state this method and the guarantees that we just obtained.

Spectral Clustering Algorithm

Input: graph G
Output: a partition of the vertices of G into two communities
1: Compute the adjacency matrix A of the graph.
2: Compute the eigenvector $v_2(A)$ corresponding to the second largest eigenvalue of A.
3: Partition the vertices into two communities based on the signs of the coefficients of $v_2(A)$. (To be specific, if $v_2(A)_j > 0$ put vertex j into first community, otherwise in the second.)

Theorem 4.5.6 (Spectral clustering for the stochastic block model)**.** *Let $G \sim G(n, p, q)$ with $p > q$, and $\min(q, p - q) = \mu > 0$. Then, with probability at least $1 - 4e^{-n}$, the spectral clustering algorithm identifies the communities of G correctly up to C/μ^2 misclassified vertices.*

Summarizing, the spectral clustering algorithm correctly classifies all except a *constant* number of vertices, provided the random graph is dense enough ($q \ge$ const) and the probabilities of within- and across-community edges are well separated ($p - q \ge$ const).

4.6 Two-sided bounds on sub-gaussian matrices

Let us return to Theorem 4.4.5, which gives an upper bound on the spectrum of an $m \times n$ matrix A with independent sub-gaussian entries. It essentially states that

$$s_1(A) \le C(\sqrt{m} + \sqrt{n})$$

with high probability. We will now improve this result in two important ways.

First, we are going to prove sharper and *two-sided* bounds on the entire spectrum of A:

$$\sqrt{m} - C\sqrt{n} \le s_i(A) \le \sqrt{m} + C\sqrt{n}.$$

In other words, we will show that a tall random matrix (with $m \gg n$) is an *approximate isometry* in the sense of Section 4.1.5.

Second, the independence of entries is going to be relaxed to just *independence of rows*. Thus we assume that the rows of A are sub-gaussian random vectors. (We studied such vectors in Section 3.4). This relaxation of independence is important in some applications to data science, where the rows of A could be samples from a high-dimensional distribution. The samples are usually independent, and so are the rows of A. But there is no reason to assume independence of columns of A, since the coordinates of the distribution (the "parameters") are usually not independent.

Theorem 4.6.1 (Two-sided bound on sub-gaussian matrices)**.** *Let A be an $m\times n$ matrix whose rows A_i are independent, mean zero, sub-gaussian isotropic random vectors in $\mathbb{R}^n$. Then for any $t \ge 0$ we have*

$$\sqrt{m} - CK^2(\sqrt{n} + t) \le s_n(A) \le s_1(A) \le \sqrt{m} + CK^2(\sqrt{n} + t) \tag{4.21}$$

with probability at least $1 - 2\exp(-t^2)$. Here $K = \max_i \|A_i\|_{\psi_2}$.

We will prove a slightly stronger conclusion than (4.21), namely that

$$\left\| \frac{1}{m} A^{\mathsf{T}} A - I_n \right\| \le K^2 \max(\delta, \delta^2) \quad \text{where} \quad \delta = C\left(\sqrt{\frac{n}{m}} + \frac{t}{\sqrt{m}}\right). \tag{4.22}$$

Using Lemma 4.1.5, one can quickly check that (4.22) indeed implies (4.21). (Do this!)

Proof We will prove (4.22) using an *ε-net argument*. This will be similar to the proof of Theorem 4.4.5, but we now use Bernstein's concentration inequality instead of Hoeffding's.

Step 1: Approximation. Using Corollary 4.2.13, we can find an $\frac{1}{4}$-net $\mathcal{N}$ of the unit sphere S^{n-1} with cardinality

$$|\mathcal{N}| \le 9^n.$$

Using the result of Exercise 4.4.3, we can evaluate the operator norm in (4.22) on the $\mathcal{N}$:

$$\left\| \frac{1}{m} A^{\mathsf{T}} A - I_n \right\| \le 2 \max_{x \in \mathcal{N}} \left| \left\langle \left(\frac{1}{m} A^{\mathsf{T}} A - I_n \right) x, x \right\rangle \right| = 2 \max_{x \in \mathcal{N}} \left| \frac{1}{m} \|Ax\|_2^2 - 1 \right|.$$

To complete the proof of (4.22) it suffices to show that, with the required probability,

$$\max_{x \in \mathcal{N}} \left| \frac{1}{m} \|Ax\|_2^2 - 1 \right| \le \frac{\varepsilon}{2} \quad \text{where} \quad \varepsilon := K^2 \max(\delta, \delta^2).$$

Step 2: Concentration. Fix $x \in S^{n-1}$ and express $\|Ax\|_2^2$ as a sum of independent random variables:

$$\|Ax\|_2^2 = \sum_{i=1}^m \langle A_i, x\rangle^2 =: \sum_{i=1}^m X_i^2 \tag{4.23}$$

where A_i denote the rows of A. By assumption, A_i are independent, isotropic, and sub-gaussian random vectors with $\|A_i\|_{\psi_2} \le K$. Thus $X_i = \langle A_i, x\rangle$ are independent sub-gaussian random variables with $\mathbb{E}\, X_i^2 = 1$ and $\|X_i\|_{\psi_2} \le K$. Therefore $X_i^2 - 1$ are independent, mean zero, and sub-exponential random variables with

$$\|X_i^2 - 1\|_{\psi_1} \le CK^2.$$

(Check this; we did a similar computation in the proof of Theorem 3.1.1.) Thus we can use Bernstein's inequality (Corollary 2.8.3) and obtain

$$\begin{aligned}
\mathbb{P}\left\{\left|\frac{1}{m}\|Ax\|_2^2 - 1\right| \ge \frac{\varepsilon}{2}\right\} &= \mathbb{P}\left\{\left|\frac{1}{m}\sum_{i=1}^m X_i^2 - 1\right| \ge \frac{\varepsilon}{2}\right\} \\
&\le 2\exp\left[-c_1 \min\left(\frac{\varepsilon^2}{K^4}, \frac{\varepsilon}{K^2}\right)m\right] \\
&= 2\exp\left[-c_1\delta^2 m\right] \quad \left(\text{since } \frac{\varepsilon}{K^2} = \max(\delta, \delta^2)\right) \\
&\le 2\exp\left[-c_1 C^2(n+t^2)\right].
\end{aligned}$$

The last bound follows from the definition of δ in (4.22) and using the inequality $(a+b)^2 \ge a^2 + b^2$ for $a, b \ge 0$.

Step 3: Union bound. Now we can unfix $x \in \mathcal{N}$ using a union bound. Recalling that $\mathcal{N}$ has cardinality bounded by 9^n, we obtain

$$\mathbb{P}\left\{\max_{x\in\mathcal{N}}\left|\frac{1}{m}\|Ax\|_2^2 - 1\right| \ge \frac{\varepsilon}{2}\right\} \le 9^n \cdot 2\exp\left[-c_1C^2(n+t^2)\right] \le 2\exp(-t^2)$$

if we chose the absolute constant C in (4.22) large enough. As we noted in Step 1, this completes the proof of the theorem. □

Exercise 4.6.2. ☕☕ Deduce from (4.22) that

$$\mathbb{E}\left\|\frac{1}{m}A^\mathsf{T}A - I_n\right\| \le CK^2\left(\sqrt{\frac{n}{m}} + \frac{n}{m}\right).$$

Hint: Use the integral identity from Lemma 1.2.1.

Exercise 4.6.3. ☕☕ Deduce from Theorem 4.6.1 the following bounds on the expectation:

$$\sqrt{m} - CK^2\sqrt{n} \le \mathbb{E}\, s_n(A) \le \mathbb{E}\, s_1(A) \le \sqrt{m} + CK^2\sqrt{n}.$$

Exercise 4.6.4. ☕☕☕ Give a simpler proof of Theorem 4.6.1, using Theorem 3.1.1 to obtain a concentration bound for $\|Ax\|_2$ and Exercise 4.4.4 to reduce to a union bound over a net.

4.7 Application: covariance estimation and clustering

Suppose we are analyzing some high-dimensional data, which is represented as points $X_1, \dots, X_m$ sampled from an unknown distribution in $\mathbb{R}^n$. One of the most basic data exploration tools is principal component analysis (PCA), which we discussed briefly in Section 3.2.1.

Since we do not have access to the full distribution but only to the finite sample $\{X_1, \dots, X_m\}$, we can only expect to compute the covariance matrix of the underlying distribution approximately. If we can do so, the Davis-Kahan theorem 4.5.5 would allow us to estimate the principal components of the underlying distribution, which are the eigenvectors of the covariance matrix.

So, how can we estimate the covariance matrix from the data? Let X denote the random vector drawn from the (unknown) distribution. Assume for simplicity that X has zero mean, and let us denote its covariance matrix by

$$\Sigma = \mathbb{E}\, X X^{\mathsf{T}}.$$

(Actually, our analysis will not require zero mean, in which case Σ is simply the second moment matrix of X, as we explained in Section 3.2.)

To estimate Σ, we can use the *sample covariance* matrix Σ_m that is computed from the sample $X_1, \dots, X_m$ as follows:

$$\Sigma_m = \frac{1}{m} \sum_{i=1}^{m} X_i X_i^{\mathsf{T}}.$$

In other words, to compute Σ we replace the expectation over the entire distribution ("population expectation") by the average over the sample ("sample expectation").

Since X_i and X are identically distributed, our estimate is unbiased, that is

$$\mathbb{E}\, \Sigma_m = \Sigma.$$

Then the law of large numbers (Theorem 1.3.1) applied to each entry of Σ yields

$$\Sigma_m \to \Sigma \quad \text{almost surely}$$

as the sample size m increases to infinity. This leads to the quantitative question: how large must the sample size m be to guarantee that

$$\Sigma_m \approx \Sigma$$

with high probability? For dimension reasons, we need at least $m \gtrsim n$ sample points. (Why?) And we now show that $m \asymp n$ sample points suffice.

Theorem 4.7.1 (Covariance estimation)**.** *Let X be a sub-gaussian random vector in $\mathbb{R}^n$. More precisely, assume that there exists $K \geq 1$ such that*[8]

$$\| \langle X, x \rangle \|_{\psi_2} \leq K \| \langle X, x \rangle \|_{L^2} \quad \textit{for any } x \in \mathbb{R}^n. \tag{4.24}$$

[8] Here we used the notation for the L^2 norm of random variables from Section 1.1, namely $\| \langle X, x \rangle \|_{L^2}^2 = \mathbb{E} \langle X, x \rangle^2 = \langle \Sigma x, x \rangle$.

Then, for every positive integer m, we have

$$\mathbb{E}\,\|\Sigma_m - \Sigma\| \le CK^2\left(\sqrt{\frac{n}{m}} + \frac{n}{m}\right)\|\Sigma\|.$$

Proof Let us first bring the random vectors $X, X_1, \ldots, X_m$ to the isotropic position. (This is only possible if Σ is invertible; think how to modify the argument in the general case.) There exist independent isotropic random vectors $Z, Z_1, \ldots, Z_m$ such that

$$X = \Sigma^{1/2} Z \quad \text{and} \quad X_i = \Sigma^{1/2} Z_i.$$

(We checked this in Exercise 3.2.2.) The assumption (4.24) then implies that

$$\|Z\|_{\psi_2} \le K \quad \text{and} \quad \|Z_i\|_{\psi_2} \le K.$$

(Check!) Then

$$\|\Sigma_m - \Sigma\| = \|\Sigma^{1/2} R_m \Sigma^{1/2}\| \le \|R_m\|\,\|\Sigma\| \quad \text{where} \quad R_m := \frac{1}{m}\sum_{i=1}^m Z_i Z_i^\mathsf{T} - I_n. \tag{4.25}$$

Consider the $m \times n$ random matrix A whose rows are Z_i^T. Then

$$\frac{1}{m}A^\mathsf{T} A - I_n = \frac{1}{m}\sum_{i=1}^m Z_i Z_i^\mathsf{T} - I_n = R_m.$$

We can apply Theorem 4.6.1 for A and get

$$\mathbb{E}\,\|R_m\| \le CK^2\left(\sqrt{\frac{n}{m}} + \frac{n}{m}\right).$$

(See Exercise 4.6.2.) Substituting this into (4.25), we complete the proof. □

Remark 4.7.2 (Sample complexity)**.** Theorem 4.7.1 implies that for any $\varepsilon \in (0,1)$, we are guaranteed to have covariance estimation with a good relative error,

$$\mathbb{E}\,\|\Sigma_m - \Sigma\| \le \varepsilon\|\Sigma\|,$$

if we take a sample of size

$$m \asymp \varepsilon^{-2} n.$$

In other words, the covariance matrix can be estimated accurately by the sample covariance matrix *if the sample size m is proportional to the dimension n.*

Exercise 4.7.3 (Tail bound)**.** ☕ Our argument also implies the following high-probability guarantee. Check that for any $u \ge 0$, we have

$$\|\Sigma_m - \Sigma\| \le CK^2\left(\sqrt{\frac{n+u}{m}} + \frac{n+u}{m}\right)\|\Sigma\|$$

with probability at least $1 - 2e^{-u}$.

4.7.1 Application: clustering of point sets

We are going to illustrate Theorem 4.7.1 with an application to clustering. Like in Section 4.5, we try to identify clusters in the data. But the nature of data will be different – instead of networks, we will now be working with point sets in $\mathbb{R}^n$. The general goal is to partition a given set of points into few clusters. What exactly constitutes cluster is not well defined in data science. But common sense suggests that the points in the same cluster should tend to be closer to each other than the points taken from different clusters.

Just like we did for networks, we will design a basic probabilistic model of point sets in $\mathbb{R}^n$ with two communities, and we will study the clustering problem for that model.

Definition 4.7.4 (Gaussian mixture model)**.** Generate m random points in $\mathbb{R}^n$ as follows. Flip a fair coin; if we get heads, draw a point from $N(\mu, I_n)$, and if we get tails, from $N(-\mu, I_n)$. This distribution of points is called the Gaussian mixture model with means μ and $-\mu$.

Equivalently, we may consider a random vector

$$X = \theta\mu + g$$

where θ is a symmetric Bernoulli random variable, $g \in N(0, I_n)$, and θ and g are independent. Draw a sample $X_1, \ldots, X_m$ of independent random vectors identically distributed with X. Then the sample is distributed according to the Gaussian mixture model; see Figure 4.5 for illustration.

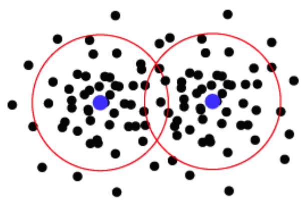


Figure 4.5 A simulation of points generated according to the Gaussian mixture model, which has two clusters with different means.

Suppose we are given a sample of m points drawn according to the Gaussian mixture model. Our goal is to identify which points belong to which cluster. To this end, we can use a variant of the *spectral clustering* algorithm that we introduced for networks in Section 3.2.1.

To see why a spectral method has a chance to work here, note that the distribution of X is not isotropic, but rather stretched in the direction of μ. (This is the horizontal direction in Figure 4.5.) Thus, we can approximately compute μ by computing the first principal component of the data. Next, we can project the data points onto the line spanned by μ, and thus classify them – just look at which side of the origin the projections lie. This leads to the following algorithm.

Spectral Clustering Algorithm

Input: points $X_1, \dots, X_m$ in $\mathbb{R}^n$

Output: a partition of the points into two clusters

1: Compute the sample covariance matrix $\Sigma_m = \frac{1}{m}\sum_{i=1}^m X_i X_i^\mathsf{T}$.

2: Compute the eigenvector $v = v_1(\Sigma_m)$ corresponding to the largest eigenvalue of Σ_m.

3: Partition the vertices into two communities based on the signs of the inner product of v with the data points. (To be specific, if $\langle v, X_i \rangle > 0$ put point X_i into the first community, otherwise in the second.)

Theorem 4.7.5 (Guarantees of spectral clustering of the Gaussian mixture model)**.** *Let $X_1, \dots, X_m$ be points in $\mathbb{R}^n$ drawn from the Gaussian mixture model as above, i.e. there are two communities with means μ and $-\mu$. Let $\varepsilon > 0$ be such that $\|\mu\|_2 \ge C\sqrt{\log(1/\varepsilon)}$. Suppose the sample size satisfies*

$$m \ge \left(\frac{n}{\|\mu\|_2}\right)^c$$

where $c > 0$ is an appropriate absolute constant. Then, with probability at least $1 - 4e^{-n}$, the Spectral Clustering Algorithm identifies the communities correctly up to εm misclassified points.

Exercise 4.7.6 (Spectral clustering of the Gaussian mixture model)**.** ☕☕☕ Prove Theorem 4.7.5 for the spectral clustering algorithm applied for the Gaussian mixture model. Proceed as follows.

(a) Compute the covariance matrix Σ of X; note that the eigenvector corresponding to the largest eigenvalue is parallel to μ.

(b) Use results about covariance estimation to show that the sample covariance matrix Σ_m is close to Σ, if the sample size m is relatively large.

(c) Use the Davis-Kahan Theorem 4.5.5 to deduce that the first eigenvector $v = v_1(\Sigma_m)$ is close to the direction of μ.

(d) Conclude that the signs of $\langle \mu, X_i \rangle$ predict well which community X_i belongs to.

(e) Since $v \approx \mu$, conclude the same for v.

4.8 Notes

The notions of covering and packing numbers and metric entropy introduced in Section 4.2 are thoroughly studied in asymptotic geometric analysis. Most of the material we covered in that section can be found in standard sources such as [11, Chapter 4] and [168].

In Section 4.3.2 we gave some basic results about error correcting codes. The book [216] offers a more systematic introduction to coding theory. Theorem 4.3.5 is a simplified version of the landmark *Gilbert-Varshamov bound* on the rate of error correcting codes. Our proof of this result relies on a bound on the binomial sum from Exercise 0.0.5. A slight tightening of the binomial sum bound leads to the following improved bound on the rate in Remark 4.3.6: there exist codes with rate

$$R \geq 1 - h(2\delta) - o(1),$$

where

$$h(x) = -x \log_2(x) + (1-x)\log_2(1-x)$$

is the *binary entropy function*. This result is known as the *Gilbert-Varshamov bound*. One can tighten up the result of Exercise 4.3.7 similarly and prove that for any error correcting code, the rate is bounded as

$$R \leq 1 - h(\delta).$$

This result is known as the *Hamming bound*.

Our introduction to non-asymptotic random matrix theory in Sections 4.4 and 4.6 mostly follows [222].

In Section 4.5 we gave an application of random matrix theory to networks. For a comprehensive introduction into the interdisciplinary area of network analysis, see e.g. the book [158]. Stochastic block models (Definition 4.5.1) were introduced in [103]. The community detection problem in stochastic block models has attracted a lot of attention: see the book [158], the survey [77], papers including [141, 230, 157, 96, 1, 27, 55, 128, 94, 108] and the references therein.

Davis-Kahan's Theorem 4.5.5, originally proved in [60], has become an invaluable tool in numerical analysis and statistics. There are numerous extensions, variants, and alternative proofs of this theorem, see in particular [226, 229, 225], [21, Section VII.3], [188, Chapter V].

In Section 4.7 we discussed covariance estimation following [222]; more general results will appear in Section 9.2.3. The covariance estimation problem has been studied extensively in high-dimensional statistics, see e.g. [222, 174, 119, 43, 131, 53] and the references therein.

In Section 4.7.1 we gave an application to the clustering of Gaussian mixture models. This problem has been well studied in statistics and computer science communities; see e.g. [153, Chapter 6] and [112, 154, 19, 104, 10, 89].